

Tutorial Week 6

Question 1: User Login System FMEA

You're a quality engineer analyzing a simple user login system for a web application. The system authenticates users via a username and password. You need to perform an FMEA to identify potential failures and prioritize risks.

1. Identify two failure modes for the login system.
2. For each failure mode:
 - Describe the **effect** on the system or user.
 - Suggest a possible **cause**.
 - Assign **Severity (S)** (1-10: 1 = minor, 10 = catastrophic), **Occurrence (O)** (1-10: 1 = rare, 10 = frequent), and **Detection (D)** (1-10: 1 = always detected, 10 = undetectable).
 - Calculate the **Risk Priority Number (RPN)** = $S \times O \times D$.

Question 2: File Upload Feature FMEA

You're assessing a file upload feature in a cloud storage app. Users upload files via a web interface, and the system saves them to a server. Perform an FMEA to evaluate risks.

1. Identify two failure modes for the file upload feature.
2. For each failure mode:
 - Describe the **effect**.
 - Suggest a possible **cause**.
 - Assign **S**, **O**, and **D** ratings (1-10 scale).
 - Calculate the **RPN**.

Question 3: Real-Time Video Streaming Service FMEA

You're a quality engineer for a real-time video streaming service (e.g., a live sports app). The critical function under analysis is "**Stream video to user device**", which handles encoding, transmission, and playback. The system processes 10,000 streams/hour during peak events. Perform an advanced FMEA with quantitative RPNs based on the following data:

- **Historical Logs:**
 - 50 stream interruptions in 1,000 hours (0.05/hour/stream).
 - 20 buffering delays in 1,000 hours (0.02/hour/stream).
 - 10 decode errors in 1,000 hours (0.01/hour/stream).
 - 5 crashes in 1,000 hours (0.005/hour/stream).
- **Detection Metrics:**

- Monitoring catches 90% of interruptions, 80% of delays, 70% of decode errors, 50% of crashes.
- **Impact Assessment:**
 - Interruption: 10% user dropout (severe).
 - Delay: 5% user dropout (moderate).
 - Decode error: 8% user dropout (significant).
 - Crash: 15% user dropout + complaints (critical).

1. Identify four failure modes for “Stream video to user device.”
2. For each failure mode:

- Describe the **effect** on users/system.
- Identify a **cause**.
- Assign quantitative **Severity (S)**, **Occurrence (O)**, and **Detection (D)** ratings (1-10 scale):
 - S: Based on user dropout % and impact (1 = <1%, 10 = >10%).
 - O: From failure rate/hour (1 = <0.001, 10 = >0.1).
 - D: From detection % (1 = >95%, 10 = <10%).
- Calculate **RPN** = $S \times O \times D$.

Question 4: E-Commerce Order Processing Failure RCA

You’re a reliability engineer performing an RCA for an e-commerce platform after a major outage where “**Order processing fails**”—customers couldn’t complete purchases during a peak sale, costing \$50K/hour. The system handles order validation, payment, and inventory updates. Using FTA, analyze the failure based on:

- **Historical Data:**
 - Network disconnect: 0.02/hour (2% chance/hour).
 - Server overload (caused by both):
 - “High load” (0.05/hour)
 - “No scaling” (0.15, 15% chance of no auto-scaling).
 - Payment gateway failure: 0.03/hour (3% chance/hour).
 - Database lock (caused by both):
 - “Concurrency bug” (0.01/hour)
 - “No lock timeout” (0.2, 20% chance of missing timeout logic).
- **System Insight:** Any one of these—network issue, server overload, payment failure, or DB lock—can halt order processing (top-level OR gate).

You are required to: -

1. Construct the fault tree.
2. Calculate the probability per hour of “Order processing fails” (round to three decimal places).
3. Calculate MTTF.

Question 5: Cloud Backup System Failure RCA

You're investigating a cloud backup system failure where **"Backup fails to complete"**, leaving customer data unprotected during a critical window. The RCA aims to trace root causes after 10% of backups failed last month (100 failures in 1,000 hours). Data includes:

- **Network Issue (caused by either):**
 - "Packet loss" (0.04/hour)
 - "Slow bandwidth" (0.03/hour).
- **Storage Failure (caused by both):**
 - "Disk full" (0.02/hour)
 - "No auto-expansion" (0.25, 25% chance of no expansion logic).
- **Software Bug (caused by either):**
 - "Compression error" (0.015/hour)
 - "File lock" (0.01/hour)
 - "No retry" (0.3, 30% chance of no retry logic).
- **Top-Level Logic:** Any one of Network, Storage, or Software failure causes the backup to fail.

Your tasks: -

1. Construct the fault tree (provided above).
2. Calculate the probability per hour of "Backup fails to complete" (round to three decimal places).
3. Calculate MTTF.